

KYLE BINDER

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APPOINTMENTS

Louisiana State University <i>RTG Postdoctoral Researcher</i>	2025 – Present
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EDUCATION

The Ohio State University <i>M.S. & Ph.D. in Mathematics</i>	2019 – 2025 Advisor: Eric Katz
Baylor University <i>B.A. in Mathematics and Linguistics (University Scholars)</i>	2015 – 2019

RESEARCH INTERESTS

Combinatorial algebraic geometry, combinatorics, toric varieties, matroids, tropical geometry.

PAPERS AND PREPRINTS

1. *Representations on the Top-Nonvanishing Weight Cohomology of a Smooth Toric Variety*, in preparation.
(~20 pp., expected October 2026)
2. *Augmented Singular Cohomology, Uniform Matroids, and Real-Rootedness* (with Lorenzo Vecchi), preprint,
[arXiv:2609.15946](https://arxiv.org/abs/2609.15946). 27 pp.
3. *The Singular Cohomology Ring of a Uniform Matroid: Combinatorics and Lefschetz Properties*, submitted,
[arXiv:2606.07490](https://arxiv.org/abs/2606.07490). 21 pp.
4. *The Singular Cohomology Ring of a Matroid*, submitted, [arXiv:2412.05732](https://arxiv.org/abs/2412.05732). 35 pp.
5. *The Unipotent Tropical Fundamental Group* (with Eric Katz), *J. London Math. Soc.* (2026), 114:e70701.
<https://doi.org/10.1112/jlms.70701>. 55 pp.
6. *Limit Sets and Internal Transitivity in Free Group Actions* (with Jonathan Meddaugh), preprint,
[arXiv:1908.07382](https://arxiv.org/abs/1908.07382). 17 pp.
7. *On 2-coloring the integers to forbid a distance and monochromatic translates of a triangle*, *Geombinatorics* **27**
(2018), no. 4, 137–146.

TALKS

<i>The Singular Cohomology Ring of a Matroid</i> University of Toronto Combinatorics Seminar	November 2026
<i>The Singular Cohomology Ring of a Matroid</i> Brown Combinatorics Seminar	November 2026
<i>The Singular Cohomology Ring of a Matroid</i> Combinatorics & Geometry BLT Seminar	September 2026
<i>A Tropical Approach to the Unipotent Fundamental Group of a Hyperplane Arrangement Complement</i> Hyperplane Arrangements and Polytopes Special Session AMS Southeastern Sectional Meeting	October 2025

<i>Cohomology of Toric Varieties from Matroids</i> LSU Geometry and Topology Seminar	September 2025
<i>The Tropical Unipotent Fundamental Group</i> 5 Minute Talk at Tropical Geometry: Moduli Spaces and Matroids Workshop	October 2024
<i>Higher Tor Groups of Matroids</i> 5 Minute Talk at Workshop of the Kentucky–Ohio Algebra Alliance (KOALA)	May 2024
<i>Higher Tor Groups of Matroids</i> OSU Combinatorics and Probability Seminar	April 2024
<i>Higher Tor Groups of Stanley–Reisner Rings of Matroids</i> 5 Minute Talk at Workshop of the Kentucky–Ohio Algebra Alliance (KOALA)	May 2023
<i>The Tropical Unipotent Fundamental Group</i> 20 Minute Talk at Workshop on Matroids and Tropical Combinatorics (MaTroCom)	January 2023
<i>A Survey of Tropical Cohomology</i> OSU Algebraic Geometry Seminar	September 2022
<i>The Geometries of Hyperplane Arrangement Complements</i> OSU Student Algebraic Geometry Seminar	April 2022
<i>p-adic Integration on Hyperelliptic Curves</i> OSU Student Algebraic Geometry Seminar	February 2021
POSTERS	
<i>The Singular Cohomology Ring of a Matroid</i> Combinatorics at the Confluence	July 2026
<i>The Singular Cohomology Ring of a Matroid</i> Texas Algebraic Geometry Symposium	April 2026
<i>The Singular Cohomology Ring of a Matroid</i> Arizona–New Mexico Symposium on Commutative Algebra	December 2025
<i>What is the Cohomology of a Non-Compact Toric Variety?</i> Kentucky, Ohio, Indiana (KOI) Combinatorics Lectures	December 2023
HONORS AND AWARDS	
OSU Distinguished University Fellowship	2019-2020, 2024-2025
Tibor Radó Research Semester	Spring 2021
Phi Beta Kappa	Fall 2018
TEACHING	
LSU Math 4023 – Applied Algebra <i>Instructor</i>	(Expected) Spring 2027
LSU Math 2065 – Elementary Differential Equations <i>Instructor</i>	(Expected) Spring 2027

LSU Math 4997 – Vertically Integrated Research: Combinatorial Topology <i>Instructor</i>	Fall 2026
LSU Math 1550 – Calculus 1 <i>Instructor</i>	Fall 2025
OSU MATH 1131 – Business Calculus <i>Recitation Instructor</i>	Spring 2024
OSU MATH 1131 – Business Calculus <i>Recitation Instructor</i>	Autumn 2022
OSU MATH 1148 – College Algebra <i>Recitation Instructor</i>	Autumn 2021
OSU MATH 1148 – College Algebra <i>Recitation Instructor; Virtual (Zoom)</i>	Autumn 2020

MENTORSHIP

Directed Reading Program (DRP) Mentor

1. “Basic Concepts of Algebraic Topology” with D.L. (Spring 2021)
2. “Quaternions for Computer Graphics” with S.B. (Fall 2021)
3. “An Invitation to Applied Category Theory” with J.W. (Spring 2022)
4. “Combinatorial Optimization” with G.M. (Spring 2023)
5. “The Mathematics of Chip-Firing” with C.B. (Fall 2023)
6. “The Mathematics of Chip-Firing” with E.R. (Fall 2026)

SERVICE AND ORGANIZATION

Special Session on Algebraic and Geometric Aspects of Matroid Theory at AMS Central Sectional Meeting <i>Co-organizer</i>	October 2026
LSU Geometry and Topology Seminar <i>Co-organizer</i>	2025 – Present
OSU Directed Reading Program (DRP) <i>Co-organizer</i>	2021 – 2025
Kentucky, Ohio, Indiana (KOI) Combinatorics Lectures <i>Local Organizer</i>	December 2023